



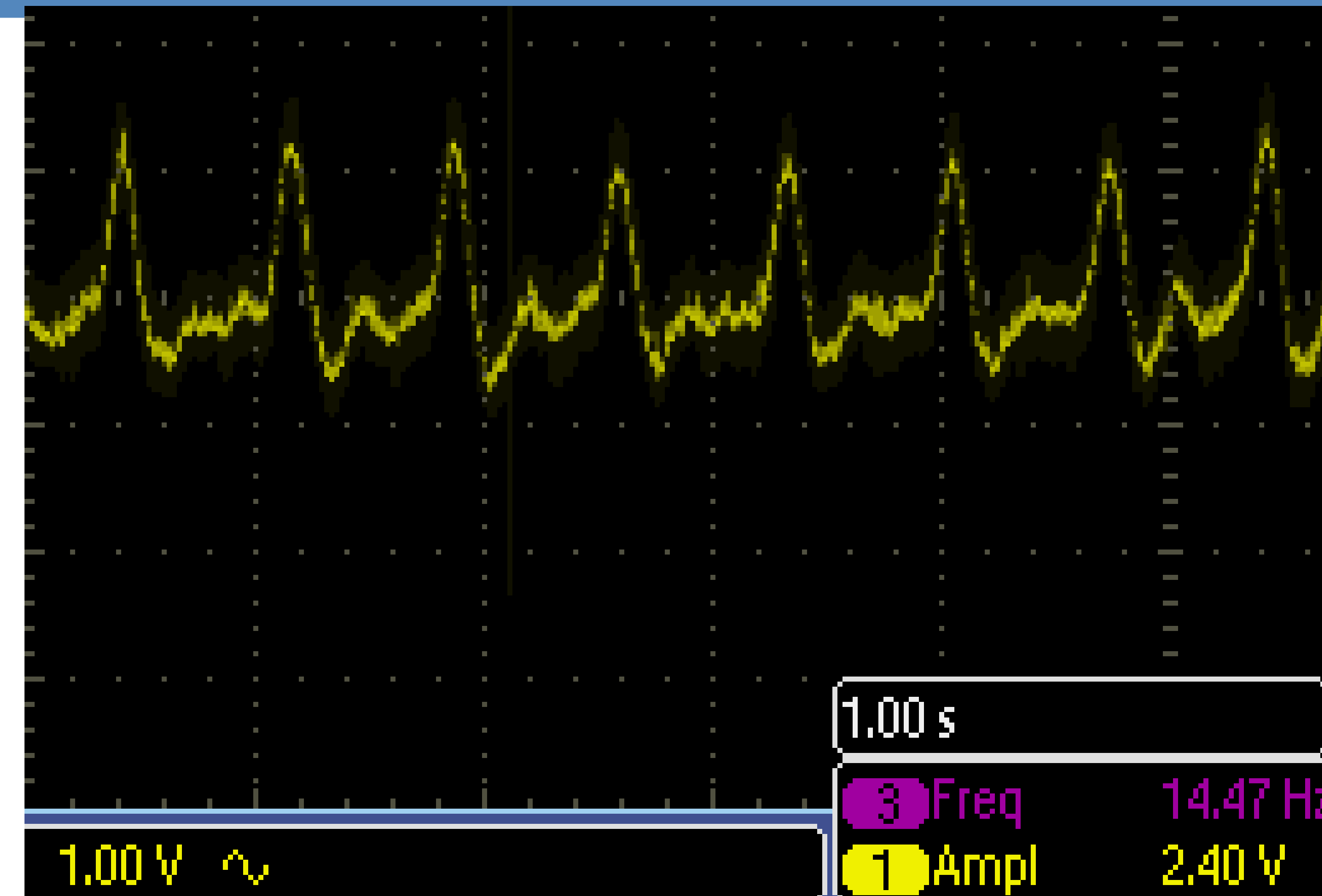
Portable Heart Rate Monitor (PHRM)

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Abstract

The Portable Heart Monitor is a wearable device that collects, processes, and displays heart-related data in real time. The system monitors important physiological signals using a pulse sensor module connected to a microcontroller. The sensor data will go through filtering and amplification to reduce noise and improve signal quality before calculating beats per minute, which is displayed to the user. The main goal of this project is to create a portable and reliable device that can monitor a person's heart rate to detect arrhythmias.

The hardware uses low-power components so the device can run efficiently on battery power and remain lightweight and portable. Along with accurate heart rate monitoring, the project will focus on maintaining stable signal processing and reliable performance in different environments. What makes this project different from many existing heart rate monitors is its focus on affordability while keeping good accuracy and functionality. By using cost-effective components and an optimized embedded system design, the device aims to provide an accessible and practical solution for everyday heart rate monitoring without sacrificing performance or portability.



Conclusion

The PHRM achieves the goal of continually monitoring user heart rate, while maintaining a compact and cost-effective design.

Future Direction:

- ◆ Battery monitoring
- ◆ Device housing
- ◆ Improve UI
- ◆ Improve irregular heart rate detection
- ◆ Sleep mode

Summary of Results

Signal Acquisition: The raw analog pulse signal from the Pulse Sensor is shown in the oscilloscope image (top right, test point TP2), confirming successful heartbeat waveform capture.

Battery Life: The device exceeds its 2-hour battery life requirement. After 2 hours of continuous monitoring, battery voltage dropped from 4.14V to 3.90V, well above the 3.08V cutoff threshold.

BPM Accuracy: Heart rate readings were validated against an Apple Watch. The PHRM stayed within ± 5 BPM under normal conditions. Occasional higher readings were observed due to pressure changes on the pulse sensor affecting the analog signal.

Arrhythmia Detection: On occasion PHRM will give a false positive. This can be caused by sensor pressure artifacts can introduce false positives, which is a known limitation of optical PPG-based detection.

